TASK 2 – DEEP LEARNING IMAGE CLASSIFICATION PROJECT

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1. INTRODUCTION

Deep Learning is a subset of Artificial Intelligence that uses neural networks to learn complex patterns from data. Image classification is one of the most common applications of deep learning where the model predicts the category of an image.

In this project, a Convolutional Neural Network (CNN) was developed to classify images using the CIFAR-10 dataset.

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2. OBJECTIVE

The main objectives of this project are:

- To understand deep learning concepts
- To build a CNN model
- To classify images into categories
- To evaluate model performance

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3. DATASET DESCRIPTION

Dataset Name:
CIFAR-10 Dataset

The dataset contains images belonging to ten categories:

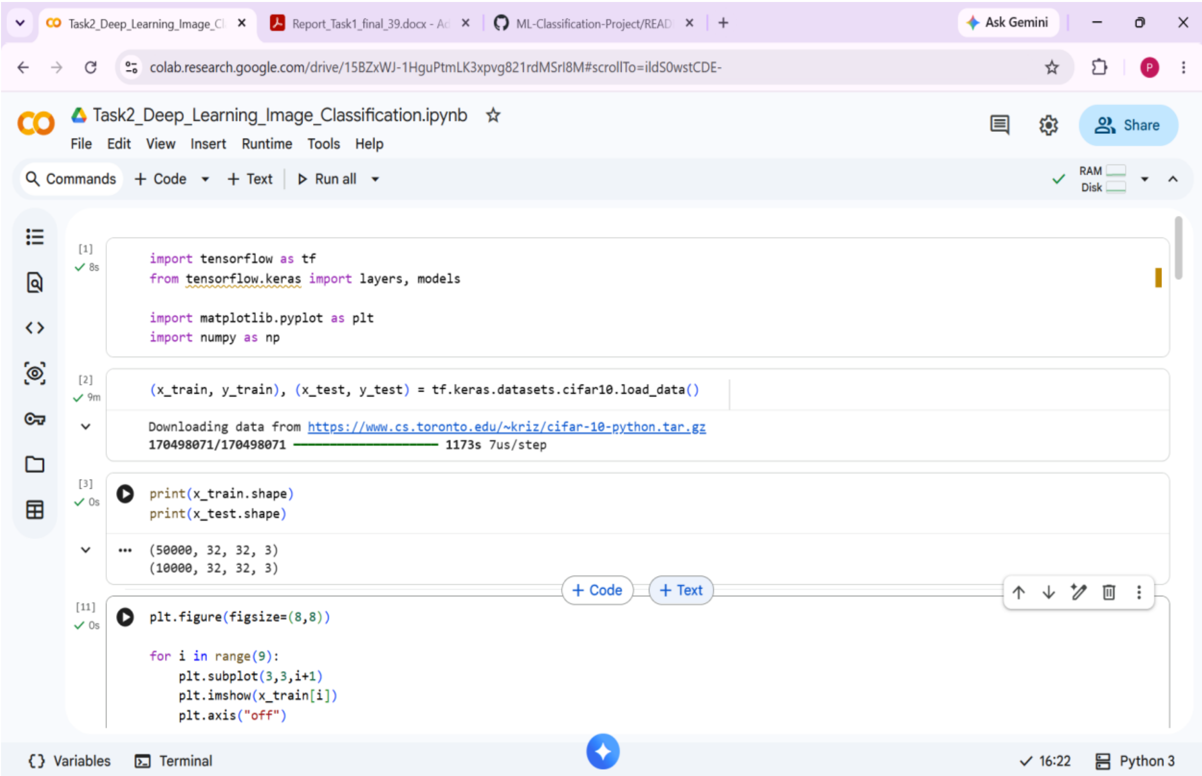
- Airplane
- Automobile
- Bird
- Cat
- Deer
- Dog
- Frog

• Horse
• Ship
• Truck

Training Images: 50,000

Testing Images: 10,000

Image Size:
32 × 32 pixels



4. DATA PREPROCESSING

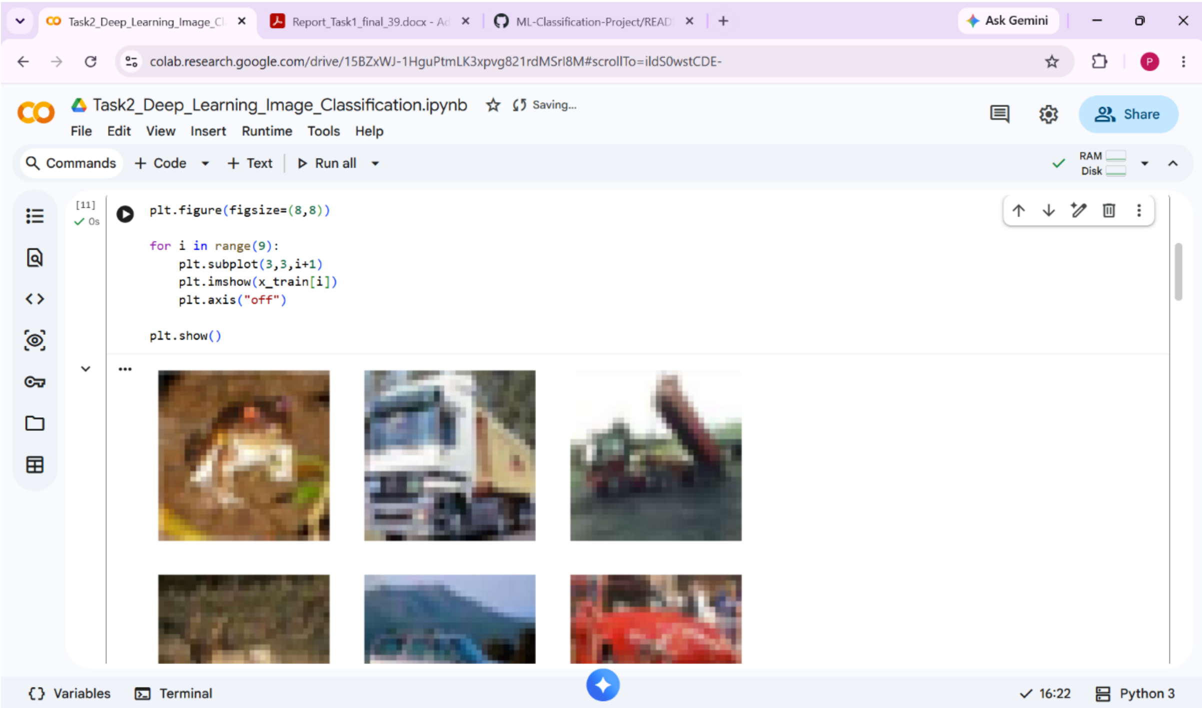
Data preprocessing techniques performed:

• Loaded dataset

• Normalized image values

• Converted image pixels between 0–1

• Split training and testing data



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5. MODEL ARCHITECTURE

The CNN model consists of:

Input Layer

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Convolution Layer

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Max Pooling Layer

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Convolution Layer

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Max Pooling Layer

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Flatten Layer

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Dense Layer

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Output Layer

6. MODEL TRAINING

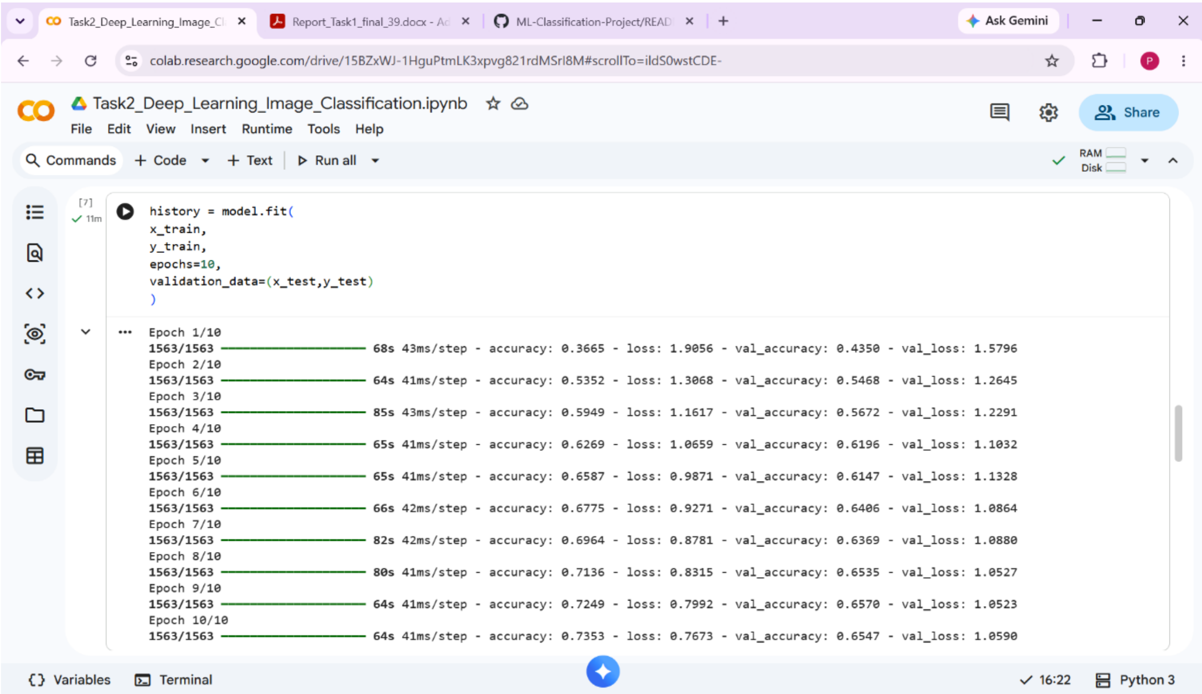
Model configuration:

Optimizer: Adam

Loss Function: Sparse Categorical Crossentropy

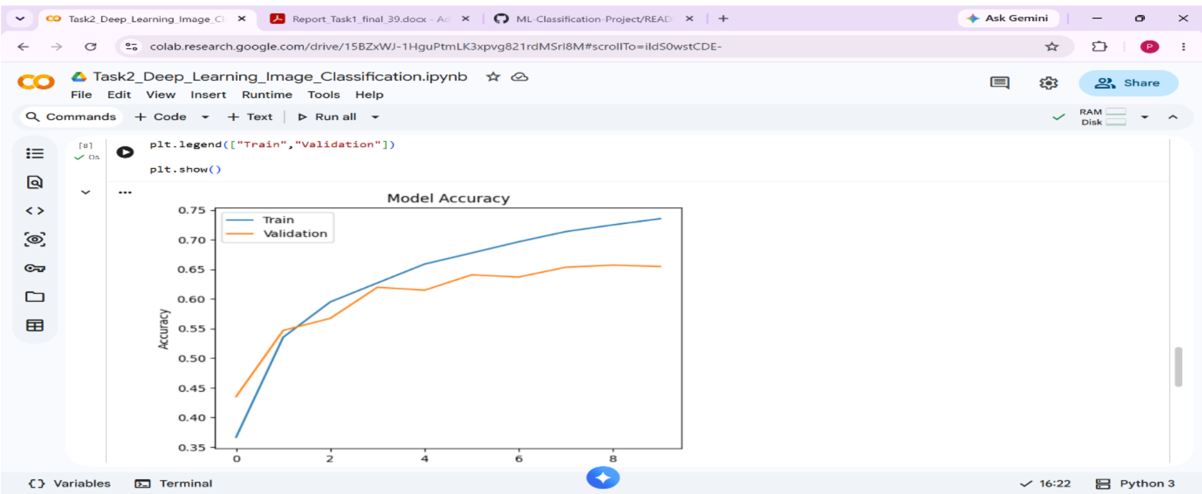
Epochs:10

Evaluation Metric: Accuracy



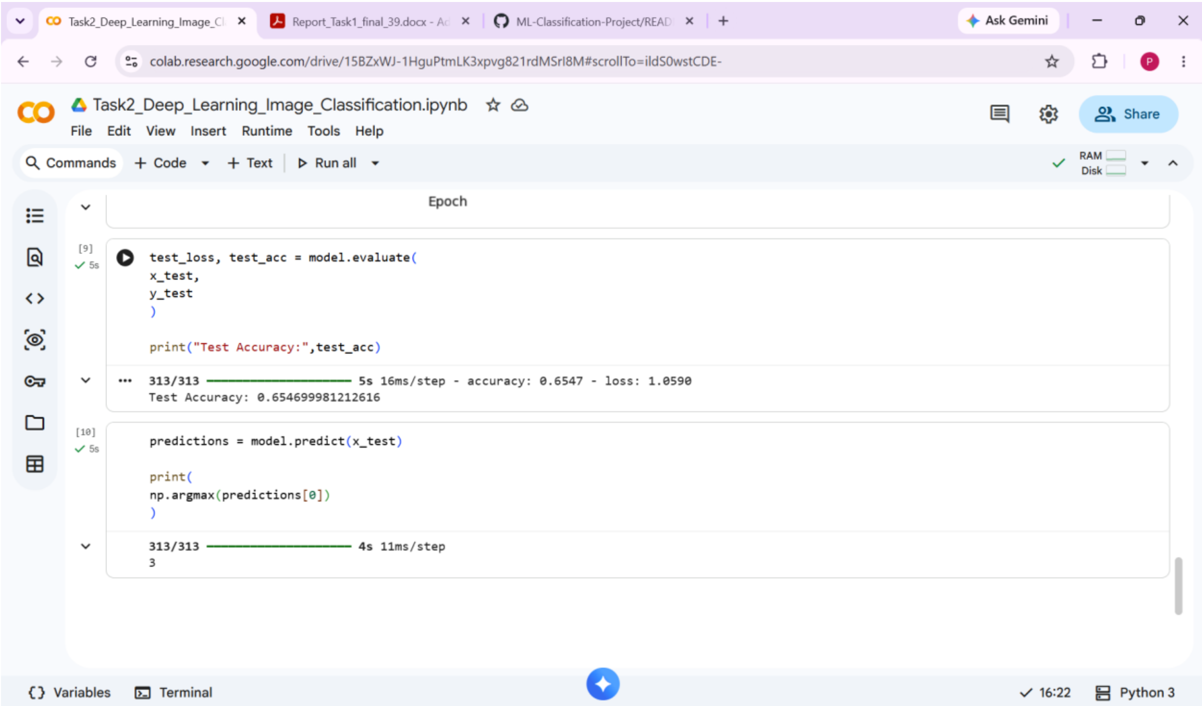
7. TRAINING CURVE

Training and validation accuracy curves were plotted to observe the model performance during training.



8. MODEL EVALUATION

The trained model was evaluated using testing data.



9. CONCLUSION

The CNN model successfully classified images into different categories using deep learning techniques. The model achieved good prediction accuracy and demonstrated the effectiveness of CNN architecture in image classification tasks.